Marine Microbial Biodiversity

# Lecture 10

## Marine Biofilms

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# Microbial Lifestyles

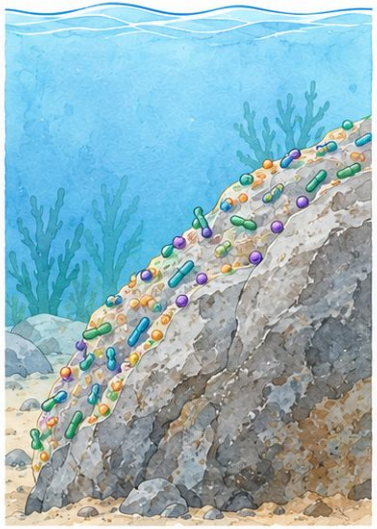
Marine prokaryotic lifestyles include several lifestyle strategies such as: i. planktonic free-floating cells (e.g., oligotrophic cyanobacteria), surface-associated biofilms and marine snow colonizers (sesto epipelon), benthic specialists, symbiotic partners, commensal or obligate pathogens, thriving through diverse metabolic pathways



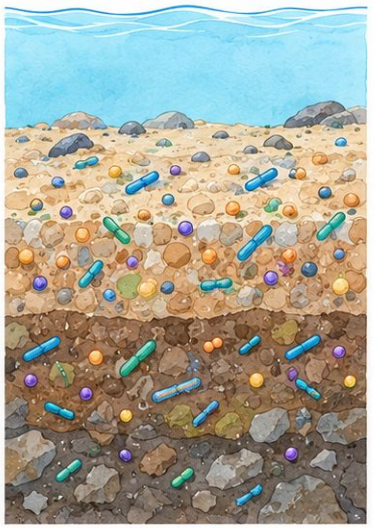
Planktonic



Seston Epipelon

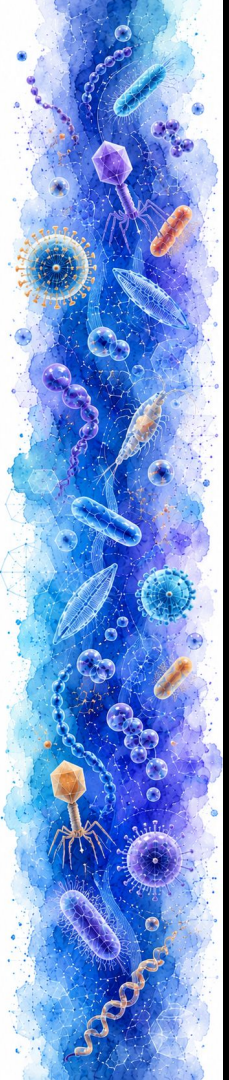


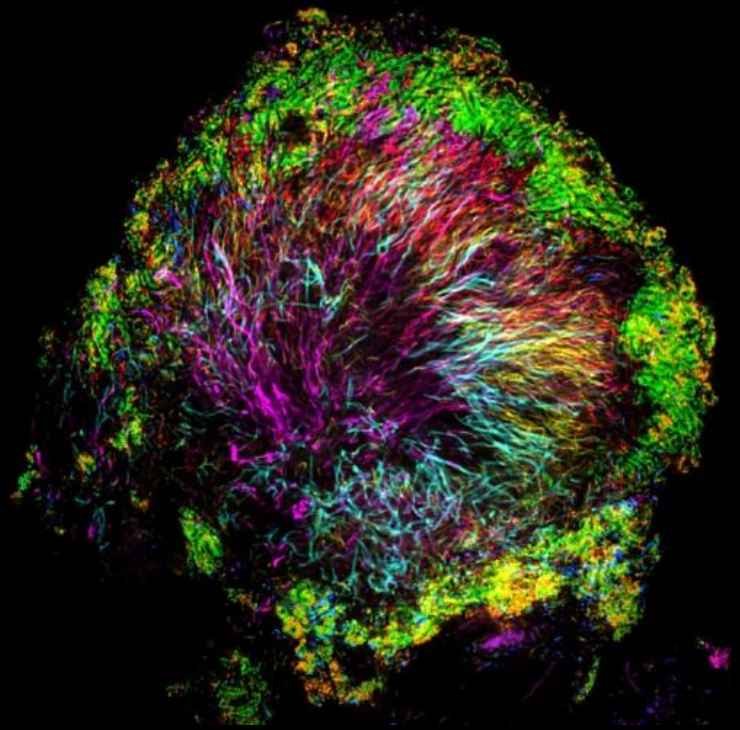
Epilithon



Episammon

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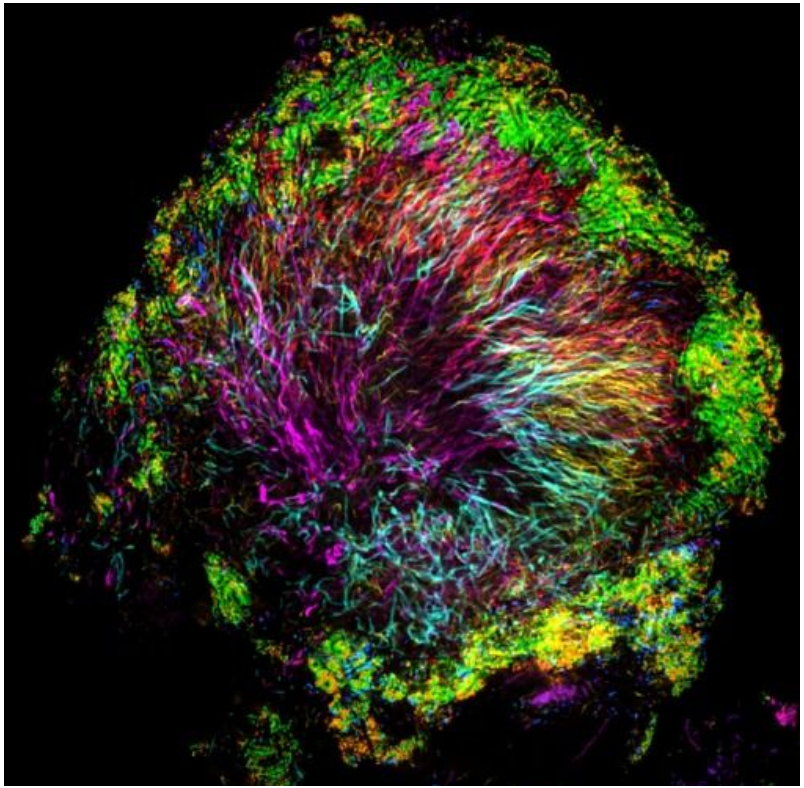


# What Is A Biofilm?

IUPAC definition of a Biofilm:
*"An aggregate of microorganisms, like bacteria, in which cells are frequently embedded within a self-produced matrix of extracellular polymeric substances (EPSs) and adhere to each other and/or to a surface"*.

Biofilm development is suggested as an ancient adaptation of prokaryotes to survive in hostile environments, and allowing colonization of new niches by active dispersal mechanisms.

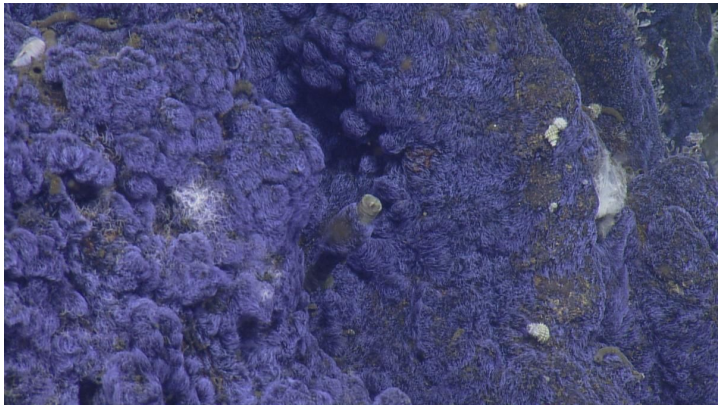
This matrix offers the bacterial community complementary protection to evade the attacks of microorganisms' antagonists or the actions of a bactericidal, such as antibiotics.



CLASI-FISH (Combinatorial Labeling and Spectral Imaging - Fluorescence in situ Hybridization)

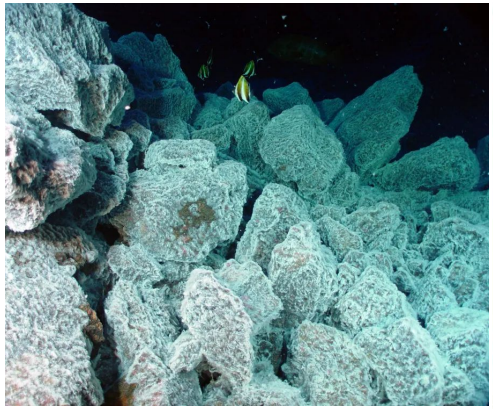
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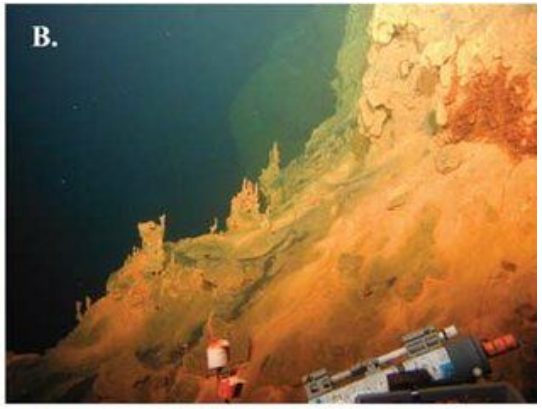
# Biofilm or Microbial Mats: A matter Of Architecture and Environmental Niche

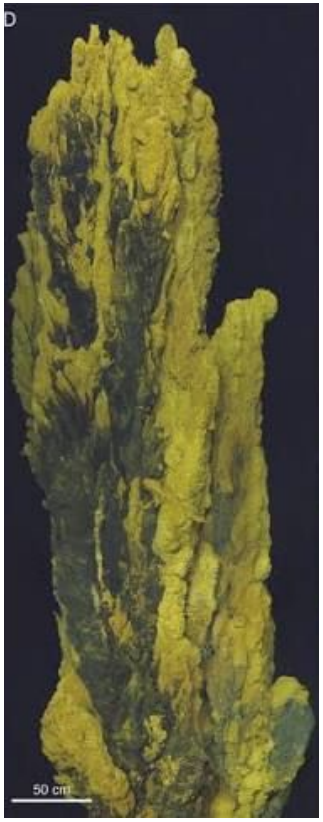






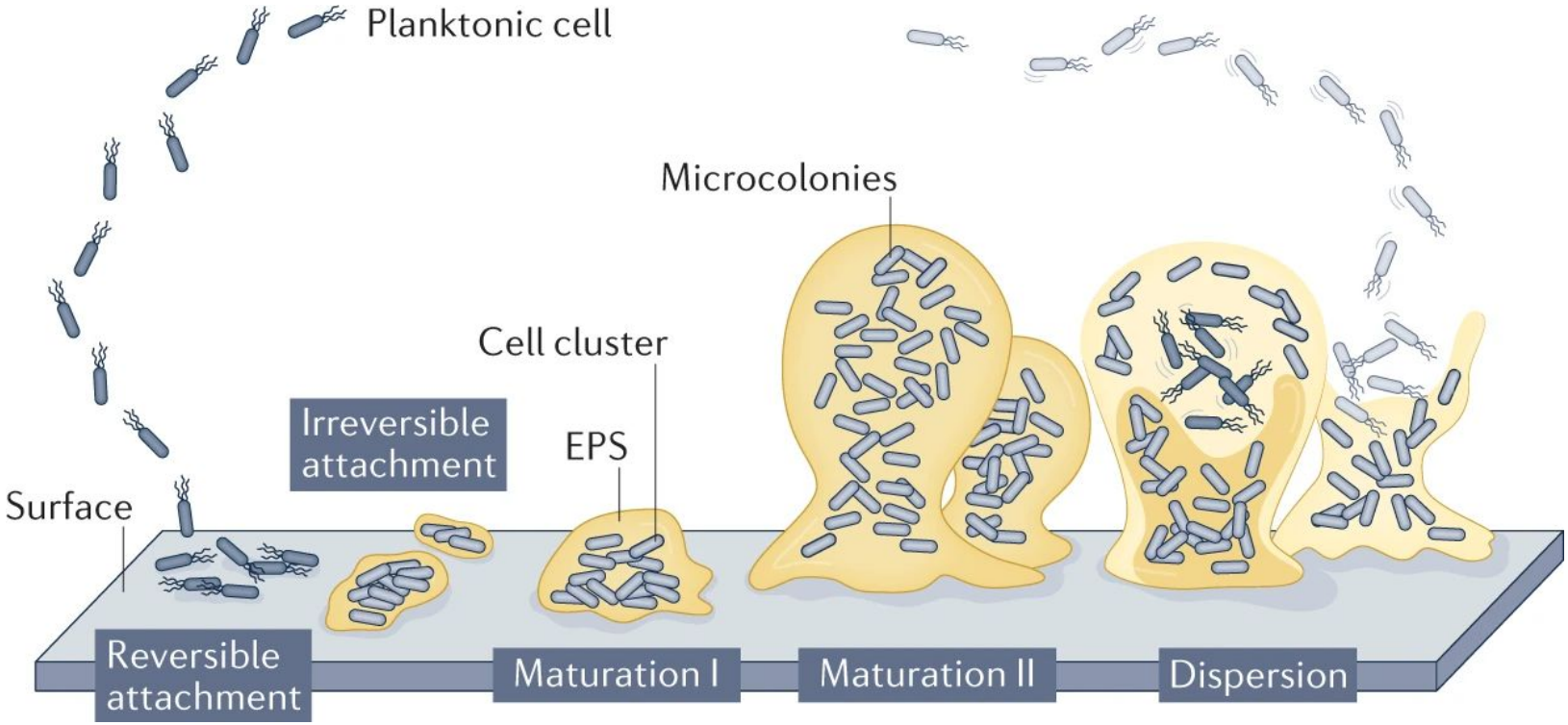





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# Biofilm Formation



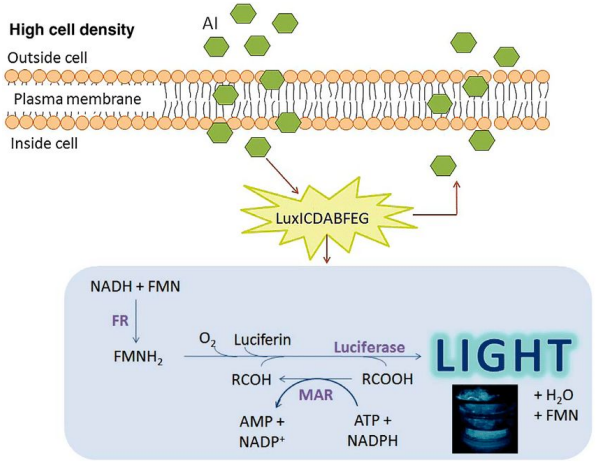

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DOI: 10.1038/s41579-022-00767-0

# Quorum-Sensing

Quorum sensing (QS) is a mechanism of intercellular communication through the controlled production, release, and detection of threshold concentrations of signal molecules. **QS is widely used by a variety of bacteria to coordinate communal behavior (e.g., biofilm formation)**. Usually, this involves the coordinated expression of certain genes in response to population density. Study of this phenomenon is leading to new insights into the physiology and ecology of attached marine microbial communities.

QS was first studied in detail in the bacterium *Vibrio fischeri*, in which it is used to control bioluminescence. When a *V. fischeri* culture reaches a certain critical density (typically late exponential, early stationary phase) bacteria synthesize a freely diffusible autoinducer (AI) molecule and release it into the medium. Accumulated AI diffuses back into the cell, binding to the protein LuxR. Its C-terminal domain binds to a palindromic sequence (lux box) upstream of the lux operon promoter, enhancing transcription and expression of the bioluminescence genes, encoding the various proteins required for the bioluminescence system.





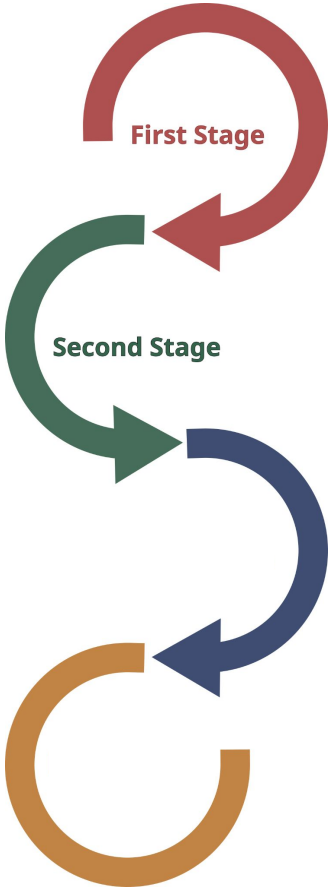


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10.1038/s41579-022-00744-7

# Biofilm Formation

First and second stage:
In the **initial attachment stage** planktonic bacterial cells attach to the surface reversibly at the pole of the cells. Physico-chemical properties such as the substrate surface roughness, hydrophobicity, film modulation, and surface charge can influence the efficiency of planktonic bacterial colonization and biofilm formation. Cell surface proteins, pili and flagella participate in initial attachment to surfaces and, in some microorganisms, are also involved in migration along the surfaces, thereby facilitating surface colonization and promoting adhesion, the irreversible stage.




- Initial adhesion of primary colonizers surfaces
- Formation of conditioning fil to facilitate adhesion

- Formation of extracellular polymeric matrix
- Irreversible adhesion of microorganisms
- Formation micro colonies

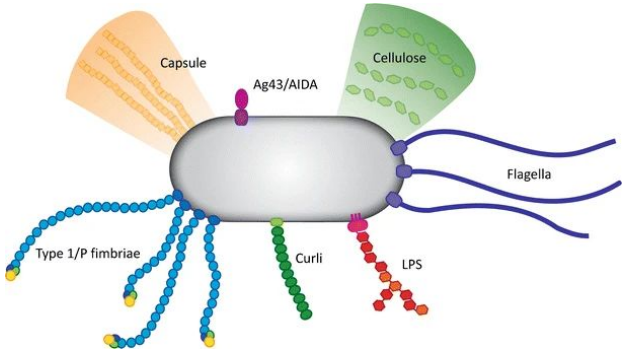
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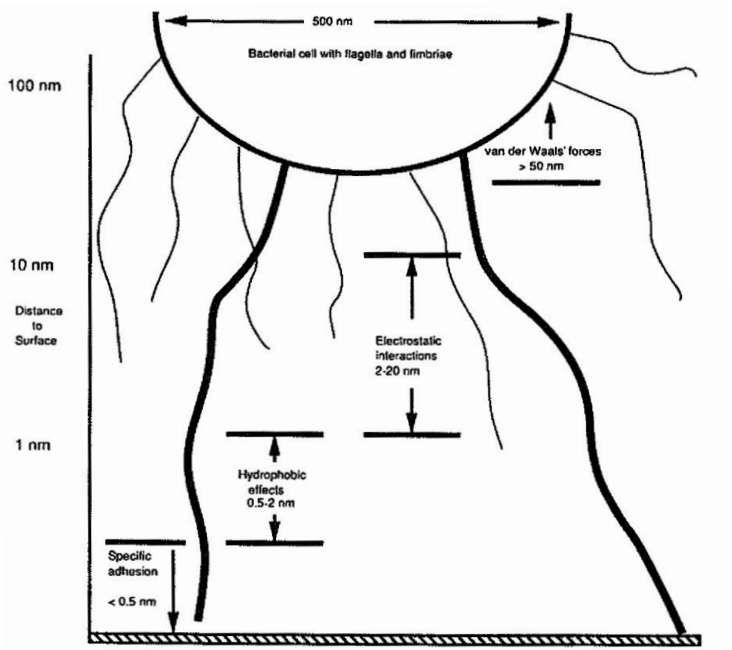
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# Adhesion Factors

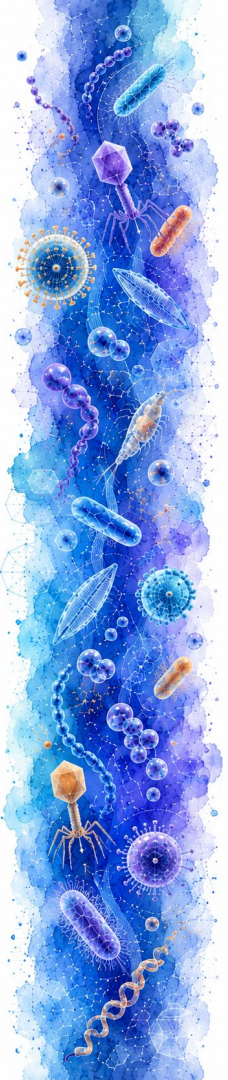
Bacterial **adherence** is the beginning of the process of colonization of a surface known as biofilm development that involves physicochemical and molecular interactions. Thus, adhesion to inert surfaces is generally related to nonspecific interactions and adhesion to biological surfaces is related to a specific ligand-receptor interaction.

**Fimbrial adhesins** also called pili form polymeric protein fibers involved in attachment to and motility on surfaces. In addition to proteinaceous adherence factors, many bacterial species produce extracellular polysaccharides involved in adhesion.



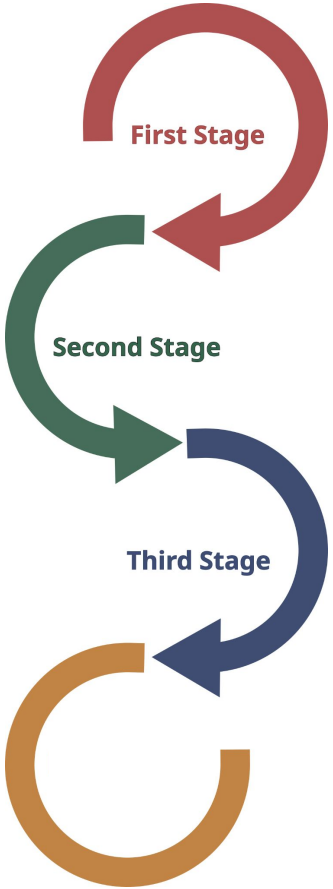

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# Biofilm Formation

Third stage:
Once attached to surfaces,
bacteria begin cell-to-cell signal communications that may control cell growth, replication, and extracellular polymeric substances (EPS) production.

EPS gradually increases within the biofilm and can accounts for 90% of the dry weight of the mature biofilm.



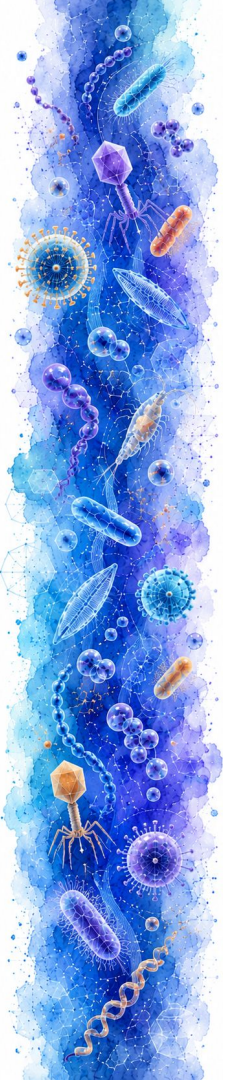

- Initial adhesion of primary colonizers surfaces
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- Formation of extracellular polymeric matrix
- Irreversible adhesion of microorganisms
- Formation micro colonies

- Biofilm maturation
- Intense microbial cell multiplication
- Water channels formation

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# Extracellular Polymeric Substance (EPS)

| EPS Components | Functions in the Context of Biofilms |
|---|---|
| Polysaccharides | Adhesion, aggregation of bacterial cells, cohesion of biofilms, protective barrier, retention of water (by hydrophilic polysaccharide), sorption of organic and inorganic compounds, sink for excess energy |
| Proteins | Enzymatic activity, electron donor or acceptor, adhesion, aggregation of bacterial cells, cohesion of biofilms, protective barrier, sorption of organic and inorganic compounds, sink for excess energy, export of cell components |
| DNA | Exchange of genetic Information, adhesion, aggregation of bacterial cells, cohesion of biofilms |
| Divalent ions | Mechanical stability, regulation of bap, regulation of EPS production |
| Surfactants and lipids | Bacterial attachment and detachment, hydrophobicity |
| Water | Provides hydrated environment, medium for movements for nutrients |

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# Biofilm Formation

Fourth stage:

## Disaggregation:

Aggregated cells, whether in suspension or associated with a surface, that shed smaller microbial aggregates or individual cells into the fluid phase.

## Erosion:

Loss of single cells or very small aggregates from the aggregate or biofilm surface due to physical forces such as fluid shear or expansion of cells by growth.

## Dispersal:

Biologically-regulated release of microbial cells from an aggregate.

## Detachment:

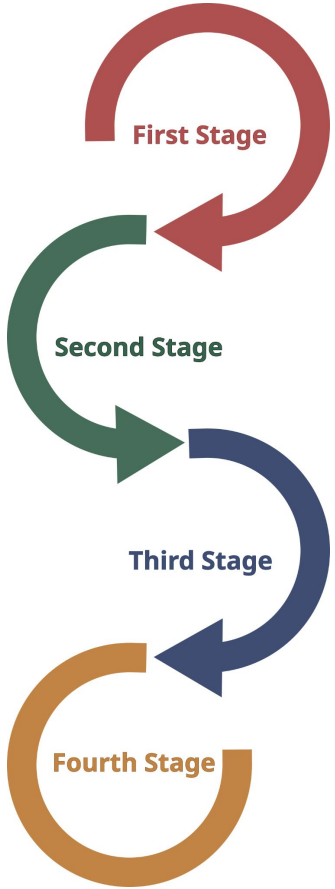
Attached microbial cells and EPS switch to a fluid-borne phase.

## Sloughing:

Release of coherent layers of surface-attached biofilm by adhesive failure (i.e., at the biofilm-substratum interface), generally by fluid shear.

## Removal:

Response to a mechanical, chemical, or enzymatic intervention that causes attached aggregates or cells to be release from the surface.




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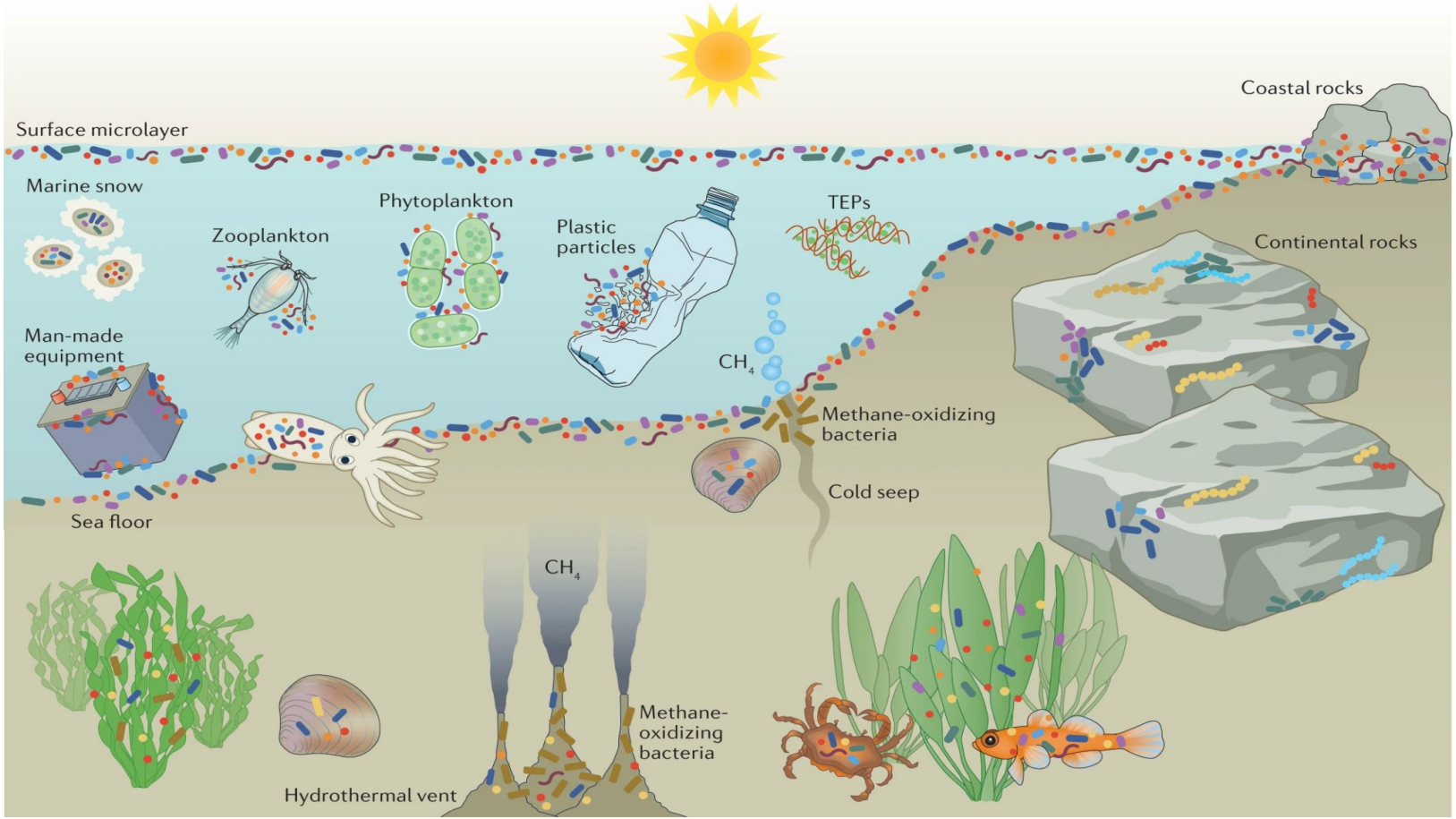
- Biofilm maturation
- Intense microbial cell multiplication
- Water channels formation

- Release of persister microbial cells from biofilm
- Adhesion of persister cells to new surfaces
- Initiation of new biofilm

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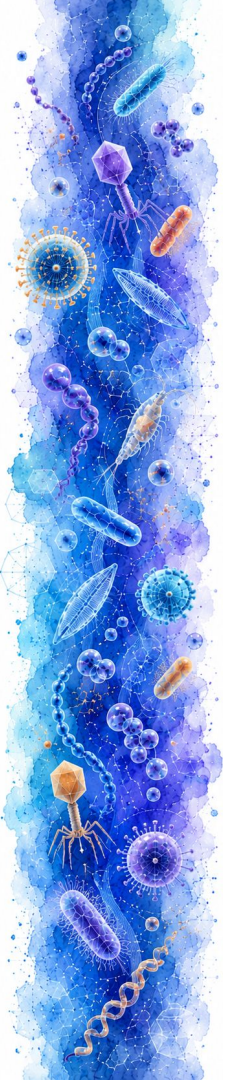
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# Biofilm In Marine Environment




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10.1038/s41579-022-00744-7



# Biofilm Key Features

. Microbial aggregates at interfaces: solid–liquid, solid–gas, liquid–liquid and liquid–gas

. Extracellular polymeric substances (EPS) matrix, mainly consisting of polysaccharides, proteins and extracellular DNA (eDNA), which forms a 'house for biofilm cells' and provides mechanical stability

. Gradients resulting in heterogeneous microenvironments in biofilms

. Wide variety of habitats supporting biodiversity and formation of synergistic micro-consortia

. Water retention and protection against dehydration

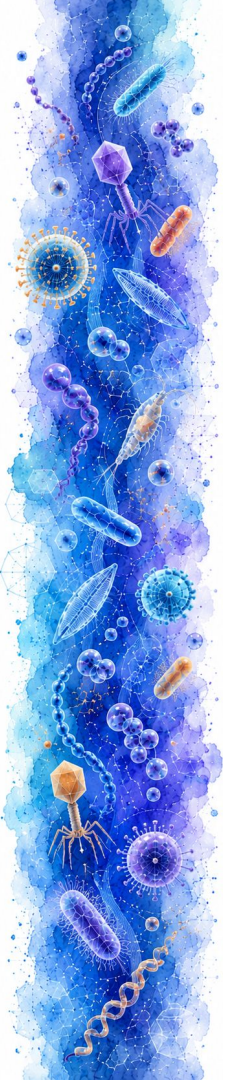
. Recycling of nutrients

. Enhanced tolerance to disinfectants, biocides and other stressors

. Enhanced intercellular communication by signalling promoting collective, coordinated behaviour, regulation of biofilm formation process

. Facilitated horizontal gene transfer (HGT) by conjugation, transduction and transformation

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# First Biofilm Observations

One of the first descriptions of biofilms, decades before the term "biofilm" was coined, involved marine bacteria. *Claude E. Zobell*, considered the "father of marine microbiology", first observed in 1936 the tendency of marine microbes present in stored seawater to adhere on bottles surface, acknowledging the importance of solid surfaces to cell growth and activity. A few years later, ZoBell showed that many bacteria found in seawater are sessile and produce a "mucilaginous holdfast".

TABLE VIII

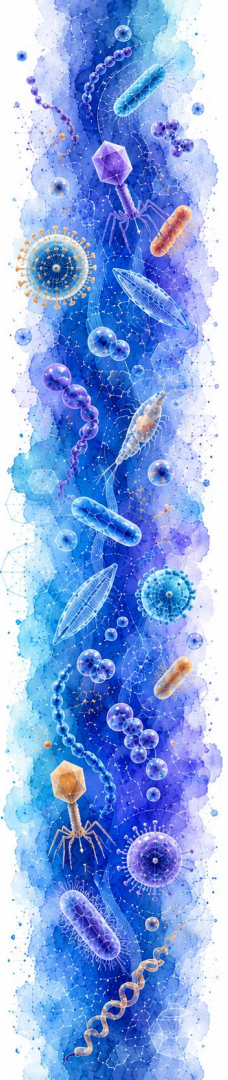
*Number of bacteria per cc. in 100 cc. of sea water poured from bottles in which it had been stored without shaking and the number per cc. in the first and second 100 cc. of sea water with which the original bottle was rinsed.*

| Storage period | Number of bacteria per cc. | | |
|---|---|---|---|
| | Original water | First rinse water | Second rinse water |
| Start | 2,800 | 0 | 0 |
| 2 days | 65,000 | 48,000 | 2,600 |
| 3 days | 108,000 | 67,000 | 3,900 |
| 4 days | 220,000 | 90,000 | 10,600 |
| 5 days | 278,000 | 122,000 | 6,450 |
| 6 days | 270,000 | 136,000 | 10,200 |
| 10 days | 10,000 | 120,000 | 9,600 |

Only in 1978 Costerton and co-workers proposed a "biofilm theory" stating that *"the majority of bacteria grow in matrix-enclosed biofilms adherent to surfaces in all nutrient-sufficient aquatic ecosystems and that these sessile bacterial cells differ profoundly from their planktonic (floating) counterparts"*.

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10.2307/1537438



# Biofilms In Marine Environment

Approximately 40-80% of living bacterial and archaeal cells on the planet reside in biofilms. Whereas some host-associated biofilms, such as those attached on human teeth, have been extensively studied, microbial diversity in marine biofilms remains largely unknown on a global scale.

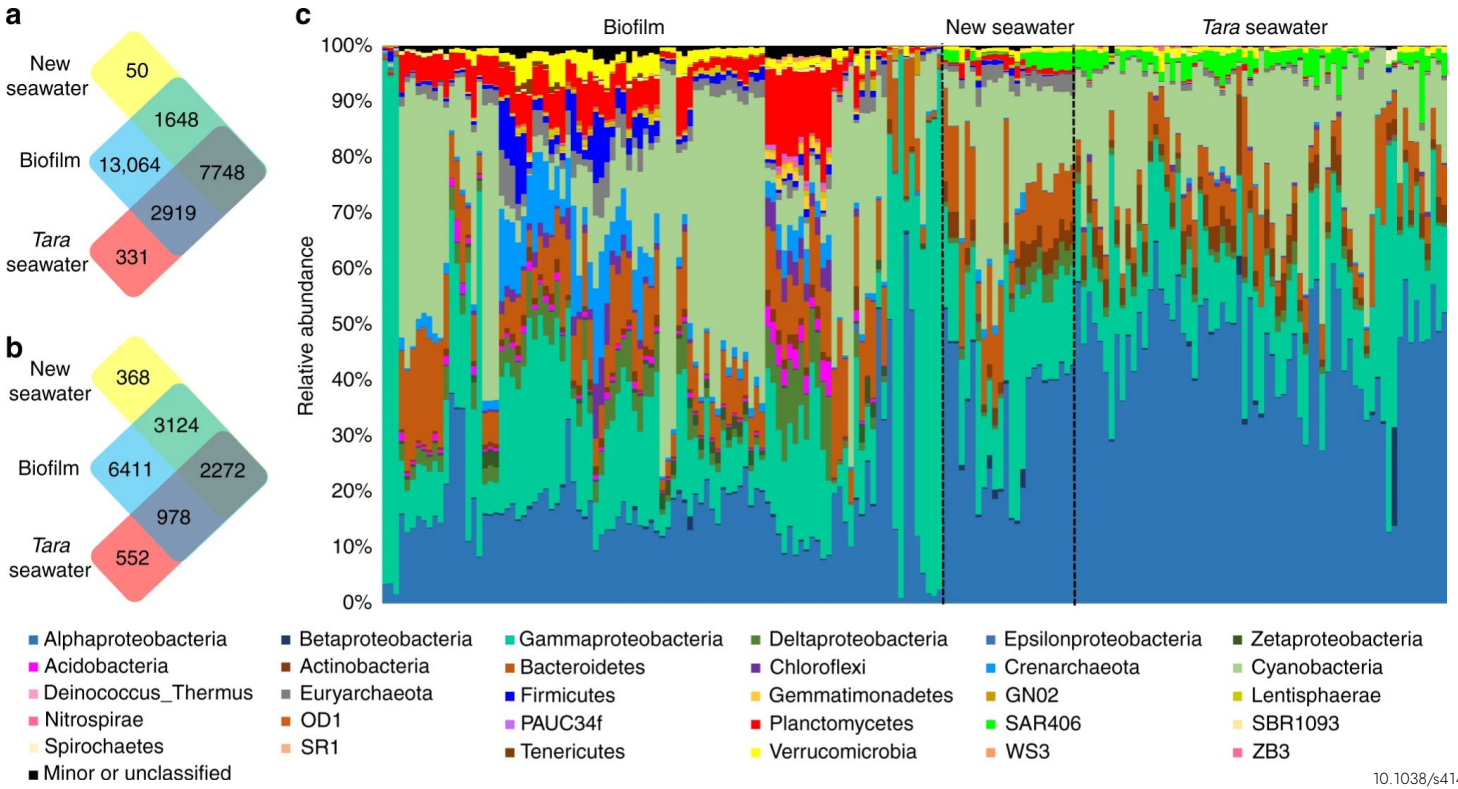
Molecular and bioinformatic techniques have greatly increased our knowledge of community structures of marine biofilms, particularly of bacteria, archaea and diatoms. The number of observed microbial taxa in natural biofilms has drastically increased **from less than 50 species identified by culture-dependent methods to about 100 operational taxonomic units (OTUs) identified by different DNA fingerprinting methods and > 10,000 OTUs by high-throughput sequencing.**

Microscopy techniques, such as **confocal laser scanning microscopy**, **confocal reflection microscopy**, **contact-angle wettability sensors**, **photoacoustic spectroscopy** and **scanning electron microscopy** have been used to monitor living organisms in marine biofilms as well as their surface properties, structures and photosynthetic ability.

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# Biofilms In Marine Environment

Comparative taxonomic analysis of the 101 biofilm and 24 seawater metagenomes sequenced as part of the current study and the 67 previously published Tara Oceans metagenomes.

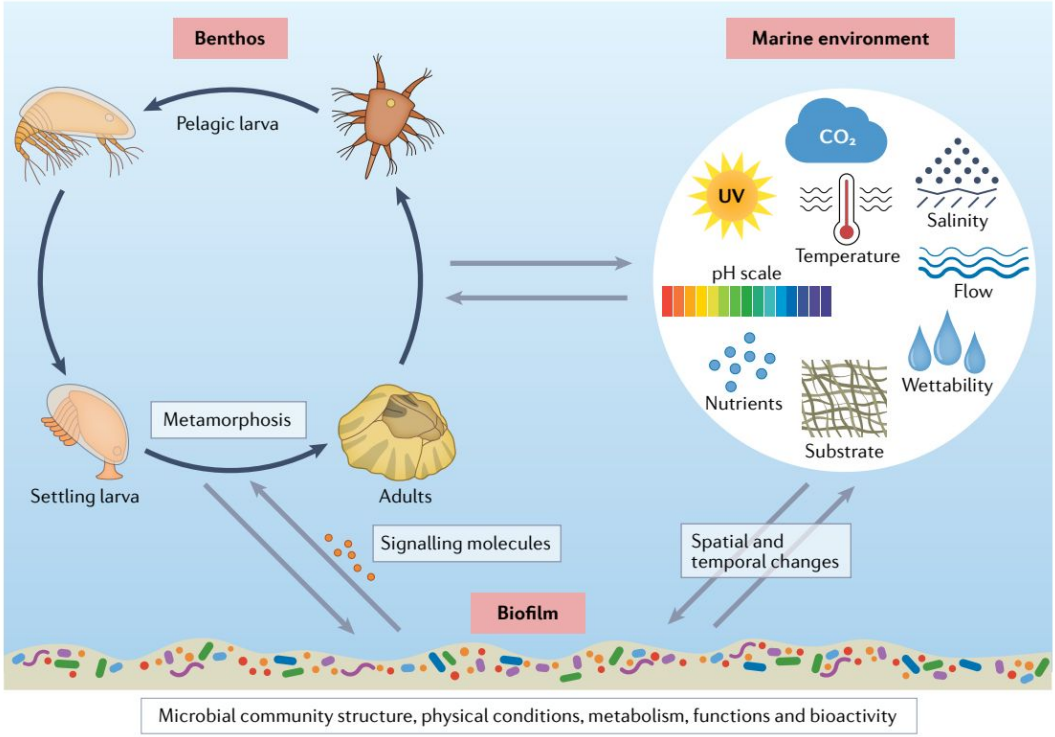



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10.1038/s41467-019-08463-z

# Biofilm Benthos Interactions

Most benthic organisms have a pelagic form (larva or spore) in their life cycle, which controls the distribution of the population across a sizeable geographic scale and locally through the selection of the optimal surface to attach to for transition to the sessile juvenile and adult form. Given that biofilms cover all marine interfaces, the interaction between settling larvae and spores and chemical and physical signals produced by biofilms determines the attachment, subsequent population dynamics, community structure, and function of benthic ecosystems.



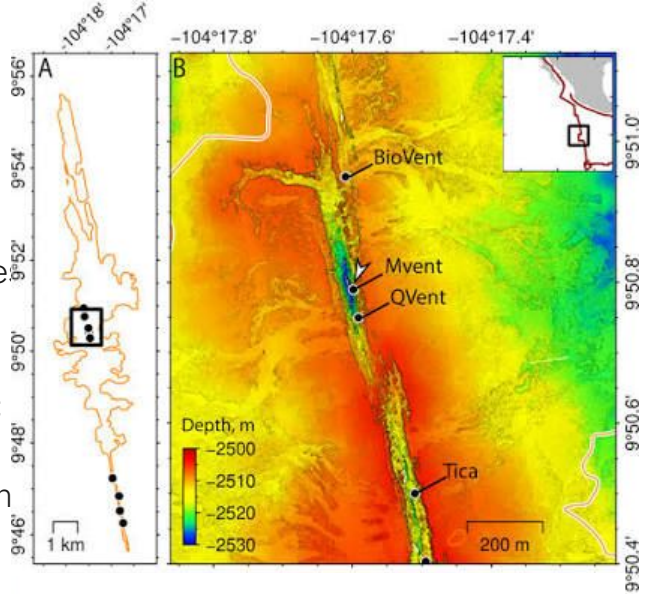

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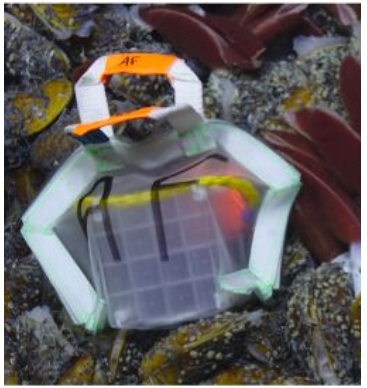
10.1038/s41579-022-00744-7

# Case study:

## Faunal colonists, including mussel settlers, respond to microbial biofilms at deep-sea hydrothermal vents

To understand the role of microbial biofilms in deep-sea ecosystem structuring, in-situ colonization experiments were conducted at Tica Vent (East Pacific Rise). Colonization surface were deployed  with either established (>1 year) or fresh biofilms across different biogenic zones for two weeks to evaluate their influence on faunal settlemen and immigration.



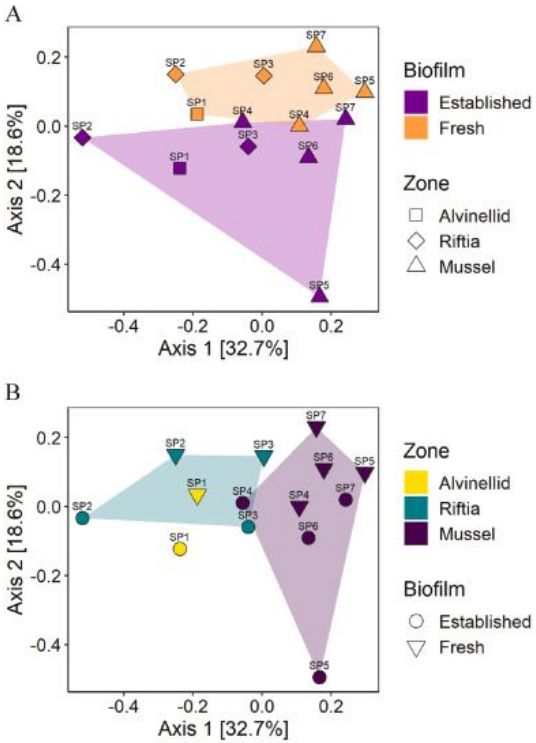


10.1016/j.dsr.2024.104314

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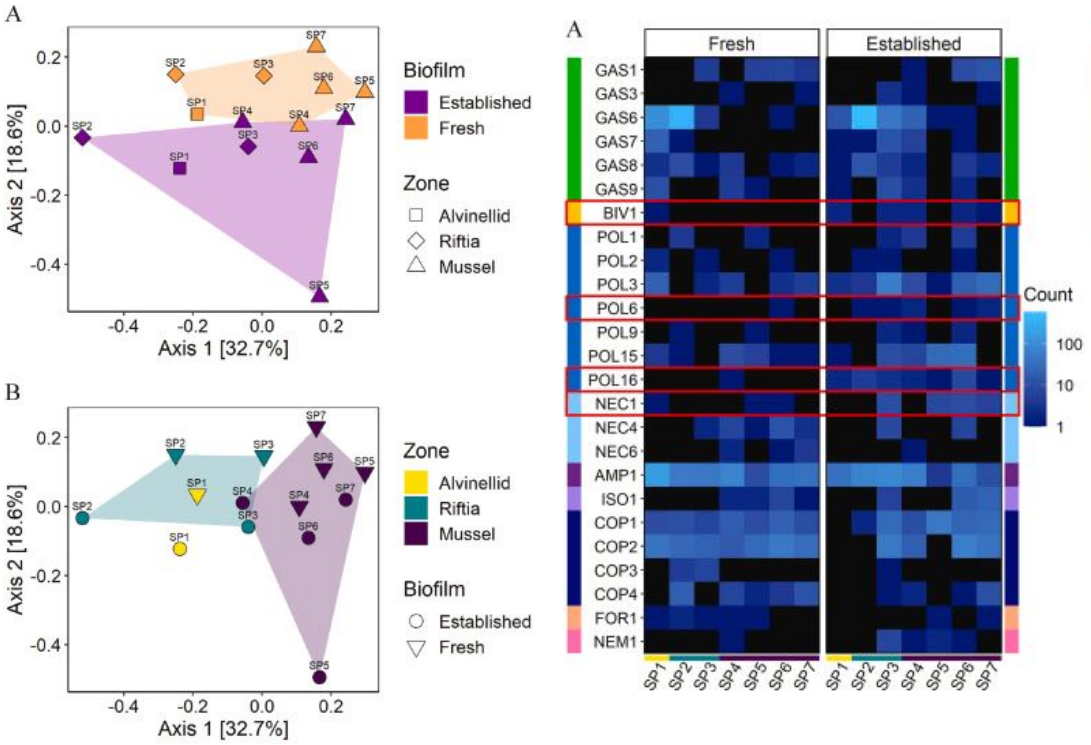



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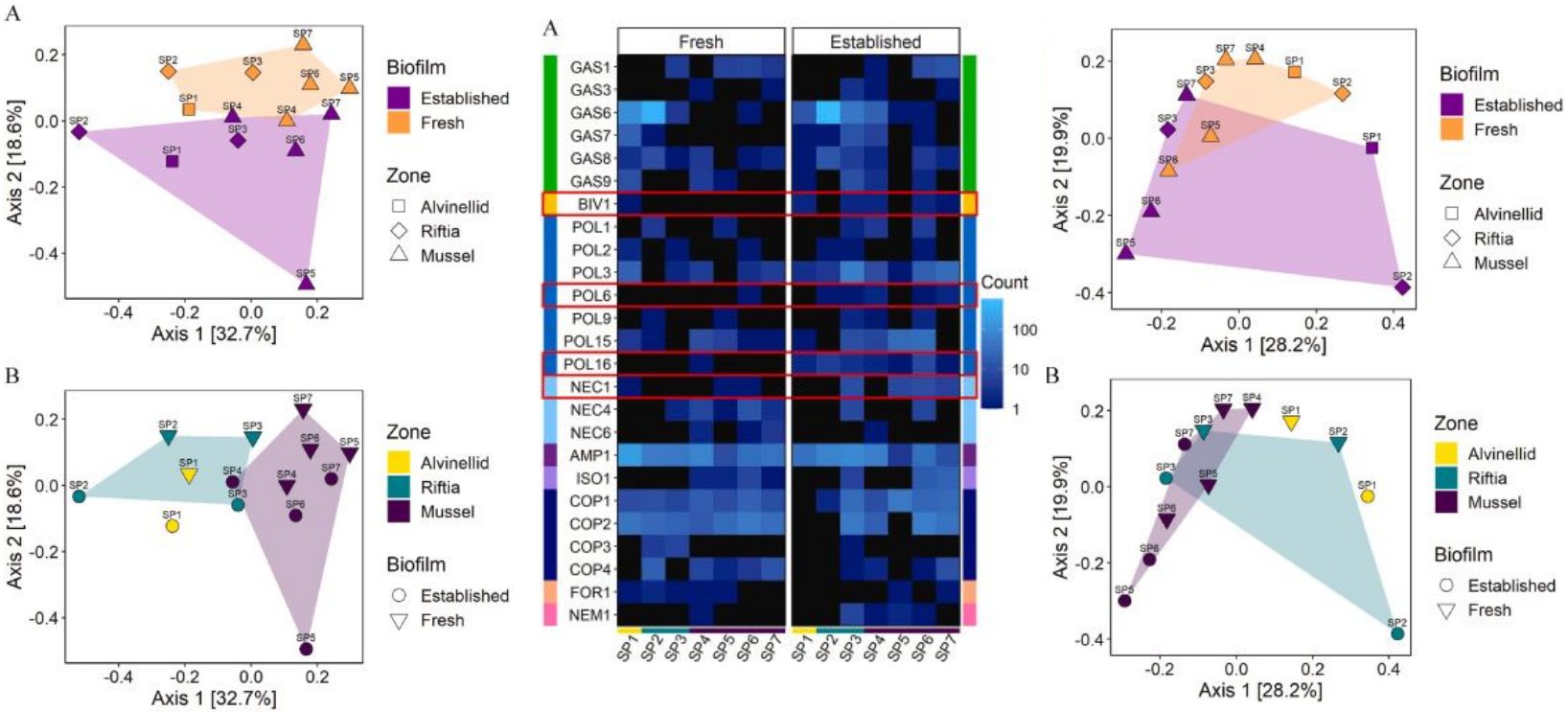



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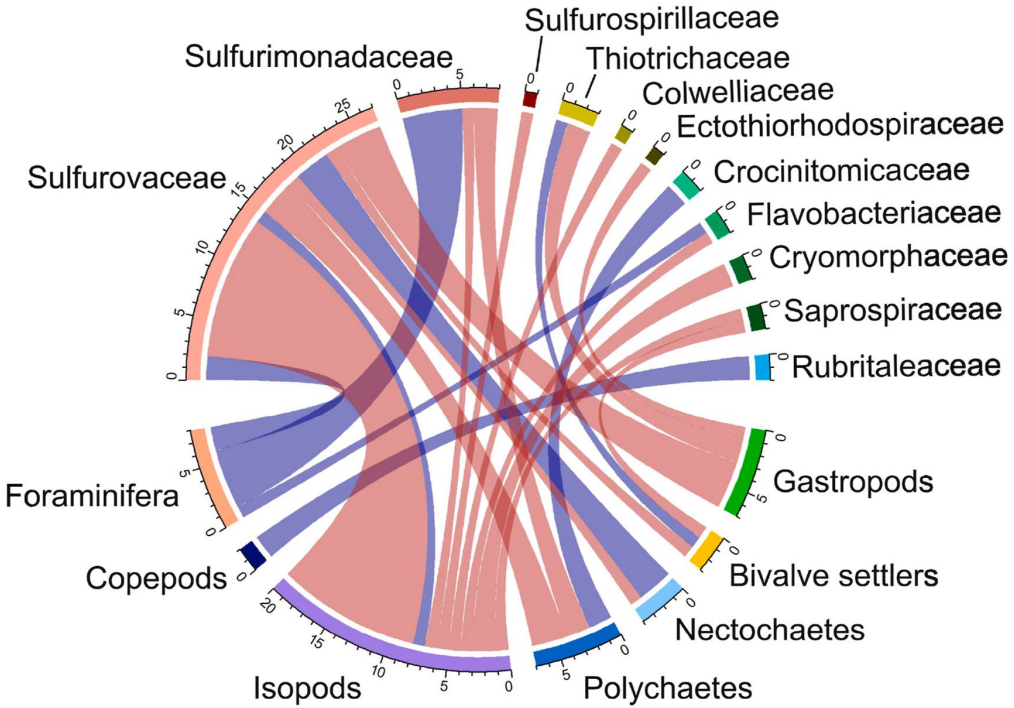

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# Case study:

## Faunal colonists, including mussel settlers, respond to microbial biofilms at deep-sea hydrothermal vents

Microbial biofilms formed near hydrothermal vents provide crucial environmental cues that govern deep-sea ecosystem recovery. Experiments at Tica Vent (East Pacific Rise) revealed that established biofilms, dominated by chemoautotrophic Campylobacterota and Gammaproteobacteria, significantly enhanced settlement and immigration of key faunal colonists, like *Bathymodiolus thermophilus*, compared to fresh biofilms.

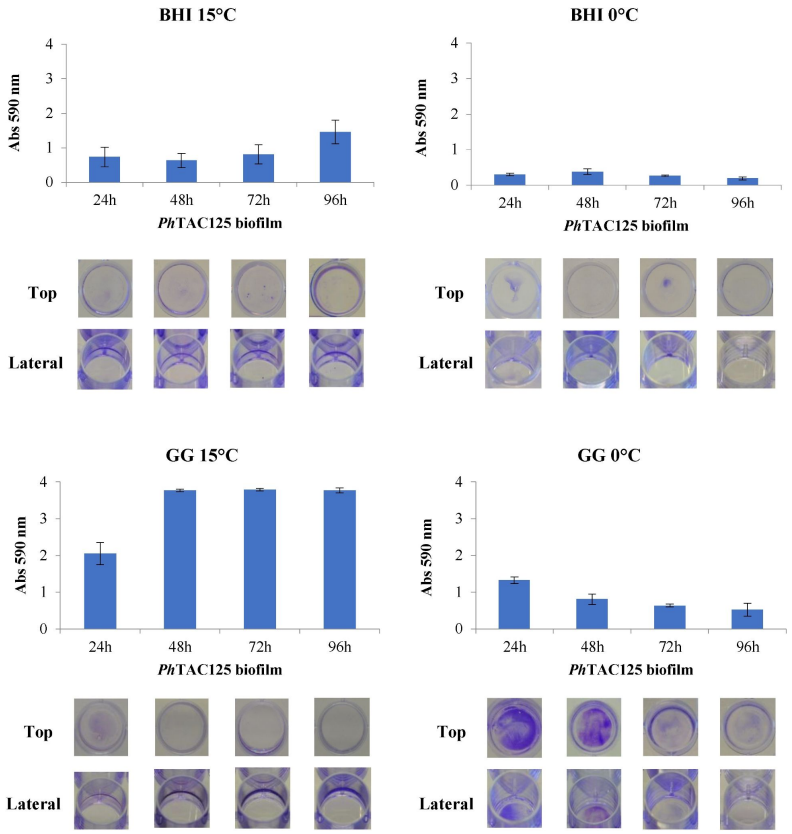



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# Case study: *Pseudoalteromonas haloplanktis* TAC125

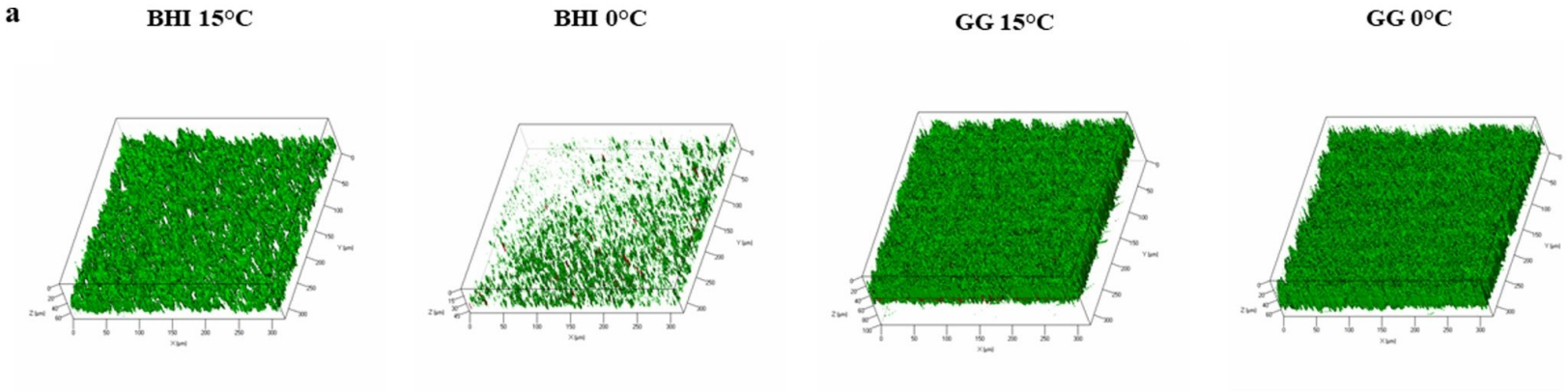
*P. haloplanktis* TAC125 is a fast growing psychroadapted Gammaproteobacterium isolated from Antarctic coastal seawater that can survive in temperatures ranging from -2,5 °C to 29 °C.
It was reported to produce an anti-biofilm agent with promising applications in medical field. But, it is able to produce this bioactive compound only if it grows in biofilm conditions. In order to understand the most suitable conditions for the anti-biofilm agent productions its biofilm formation capabilities were evaluated in different conditions.
The bacterium was grown in BHI, a rich culture medium, or in GG, a synthetic minimal medium, at 15 °C and 0 °C in static conditions, and the biofilm was evaluated at different incubation times with the crystal violet staining, and with confocal laser scanning microscopy (CLSM) technique.

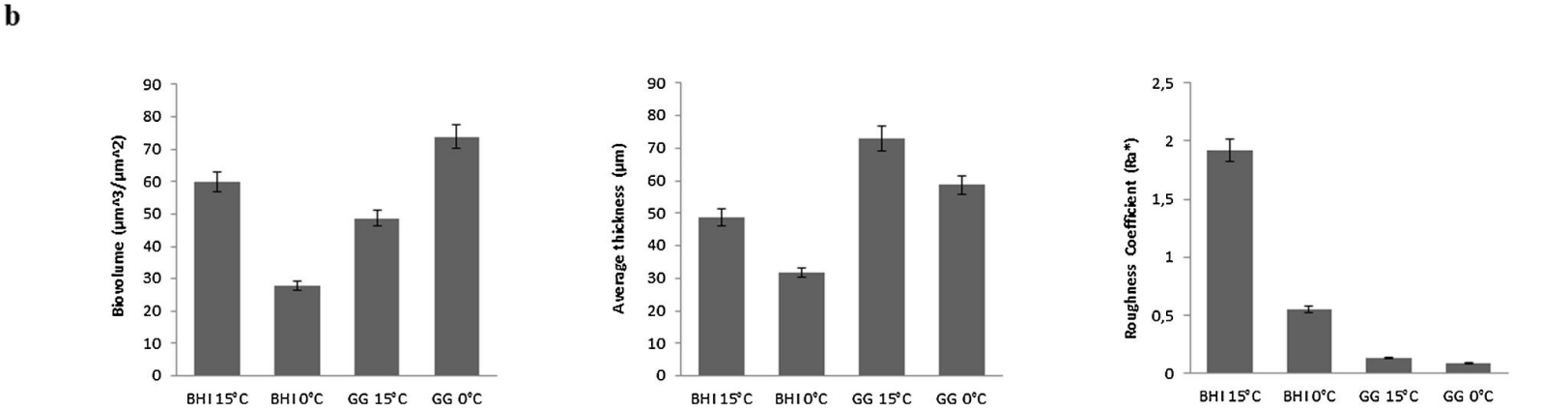



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10.1016/j.micres.2018.09.010

# Case study: *Pseudoalteromonas haloplanktis* TAC125



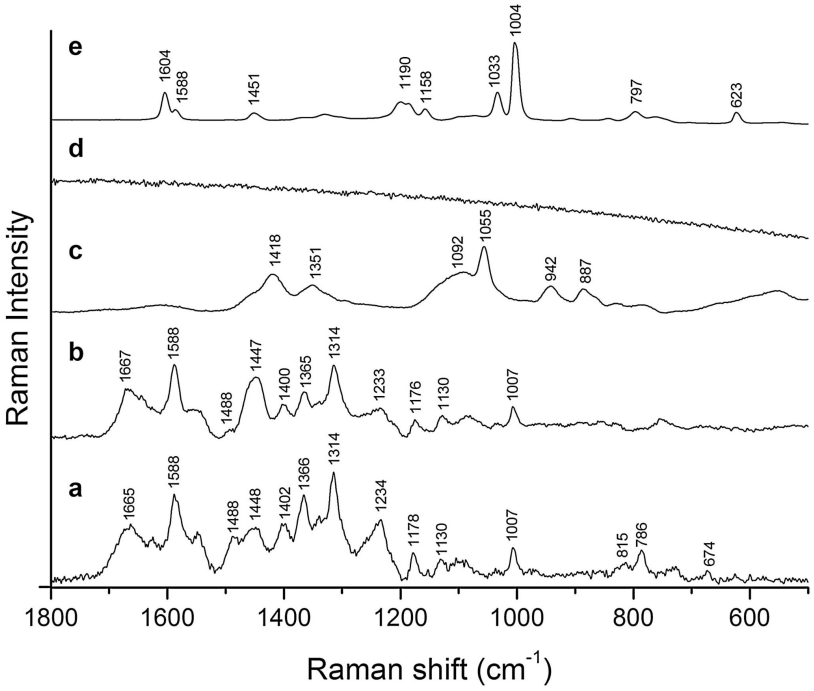



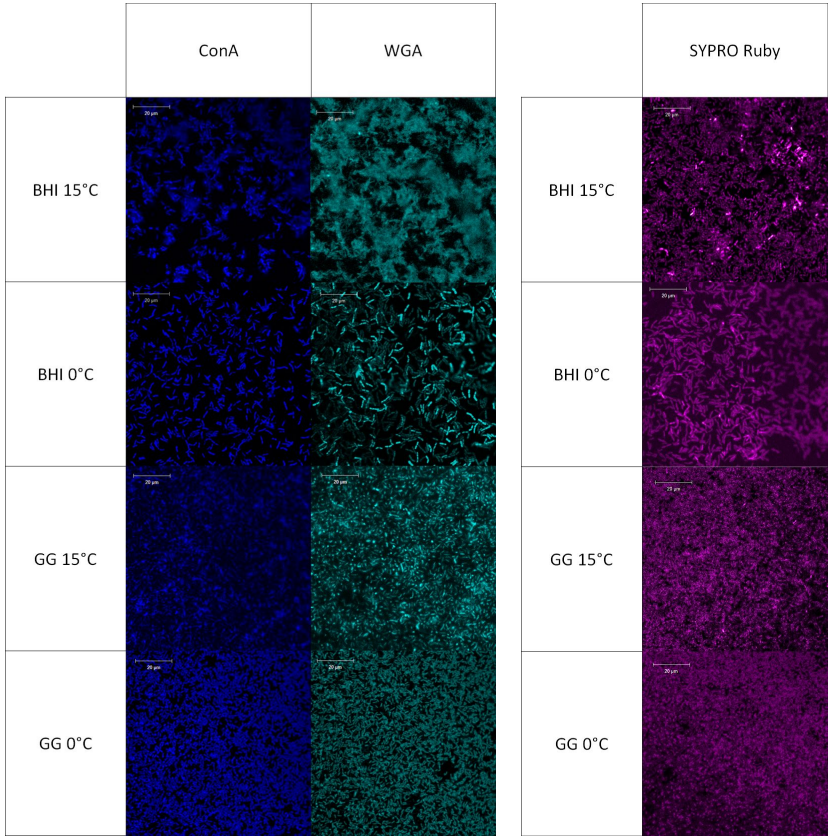

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10.1016/j.micres.2018.09.010

# Case study: *Pseudoalteromonas haloplanktis* TAC125

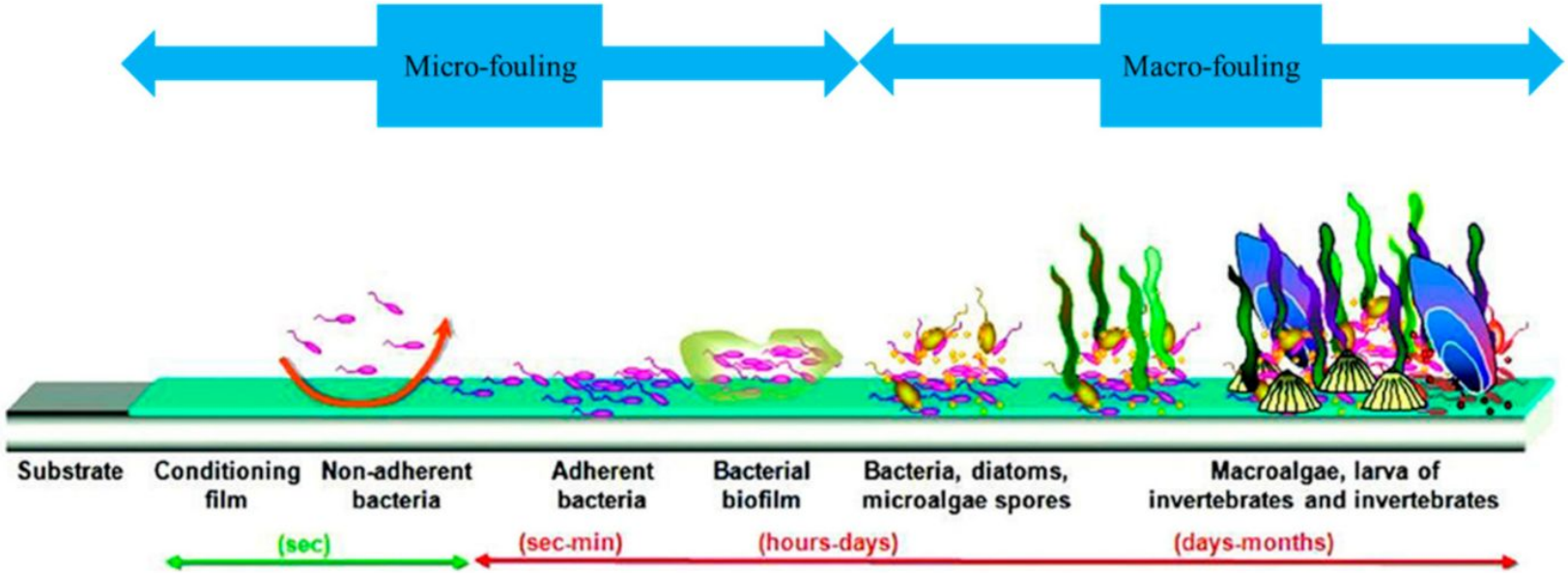
CLSM and Raman Microspectroscopy provide molecular details of the chemical composition of bacterial biofilms.




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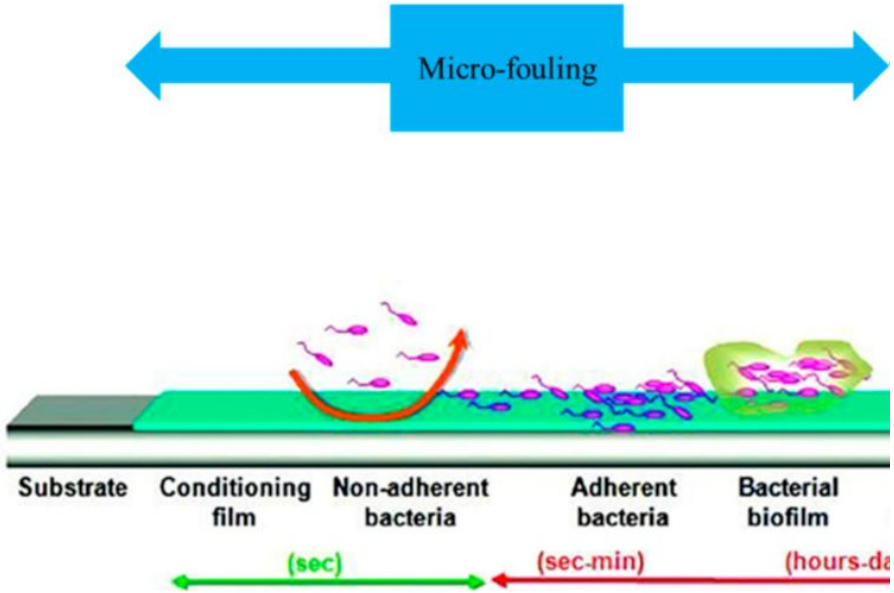
# Biofouling




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10.3390/su13052625

# Biofouling



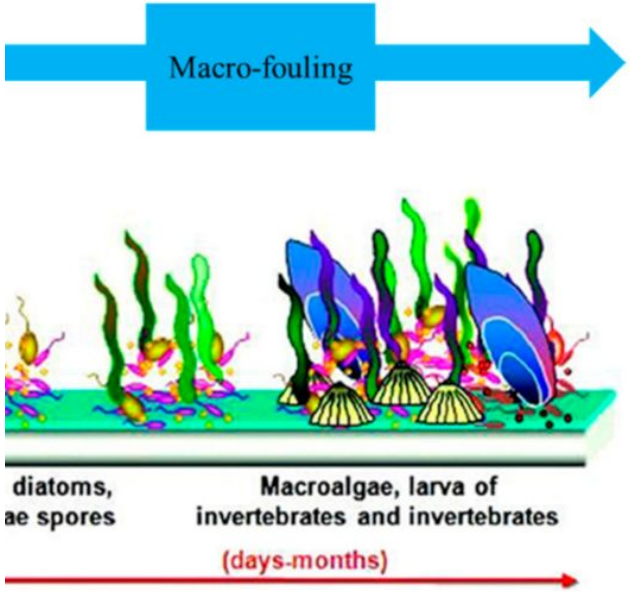

Generally, prior to bacterial adhesion, adsorption of dissolved organic and inorganic matter onto surfaces usually takes place. This is defined as **conditioning film (CF)**. The conditioning film is composed of glycoproteins, humic acids, proteins, aromatic amino acids, carbohydrates, uronic acids and/or other unspecified macromolecules. After the formation of a nutrient-rich layer of CF in the first few hours, microbial colonization follows in less than a day.

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# Biofouling

Together with diatoms and other microorganisms, bacteria are responsible for **microfouling**, allowing the adhesion of larger organisms which cause **macrofouling**. Macrofoulers are typically represented by numerous species of algae, sponges, coelenterates, polychaetes, bryozoans, molluscs, barnacles, and amphipods. These organisms derive from the water column where they are represented by larvae or spores that rely on hard substrata onto which they must settle to successfully complete their life cycles. It is in the upper layers of the oceans, especially along continental margins, that fouling communities are most abundant, although they also occur at depths of several hundred metres. The shallow water fouling communities are, in several respects, similar to naturally-occurring hard shore assemblages and consequently often mirror the competitive interactions between individuals for the acquisition of space, which is the major limiting resource in fouling habitats




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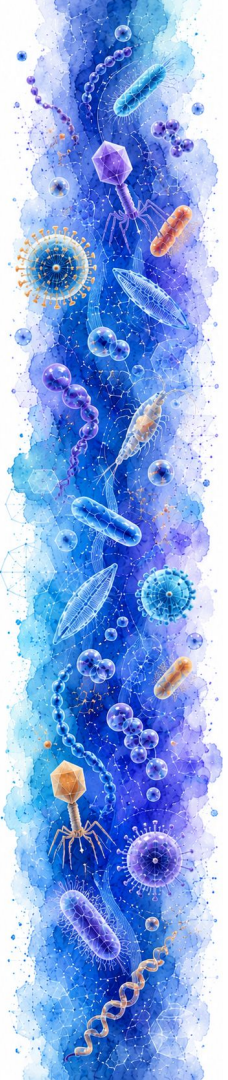
# Biofouling

## Consequences

. Increased hydrodynamic drag
. Higher fuel consumption and emissions
. Financial and regulatory penalties
. Material damage and corrosion
. Invasive species transfer




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# Biofouling Impact On Industries

The environmental impact and the costs to several industries that derive from biofouling are substantial.

**TABLE 1 |** Estimated costs due to biofilm formation in industrial activities.

| Industry | Problem | Estimated cost | References |
|---|---|---|---|
| Aquaculture | Fouling of both stock and infrastructures | 1.5–3 billion US dollars per year | Fitridge et al., 2012 |
| | Biofouling on fish cages and shellfish sites | 5–10% of the industry value in Europe (ca. 260 million euros per year) | Lane and Willemsen, 2004 |
| Heat exchange | Decreased heat transfer and induced corrosion | ca. 7.5% of maintenance costs of a process plant | Ibrahim, 2012 |
| Oil and gas industry | Microbial influenced corrosion | 20–30% of corrosion-related costs | Skovhus et al., 2017 |
| Maritime transport | Increased drag | 35–50% increased fuel consumption | Schultz et al., 2011 |
| | Fouling of ballast water with non-indigenous species | 1.6–4% of annual operational cost for a ship | Fernandes et al., 2016 |
| Water desalination | Biofouling of reverse osmosis membranes | 5–20% of operational costs for cleaning; cost of biofouling ca. 30% of total operating costs | Flemming, 1997; Maddah and Chogle, 2017 |

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# Biofouling Impact on Marine Corrosion

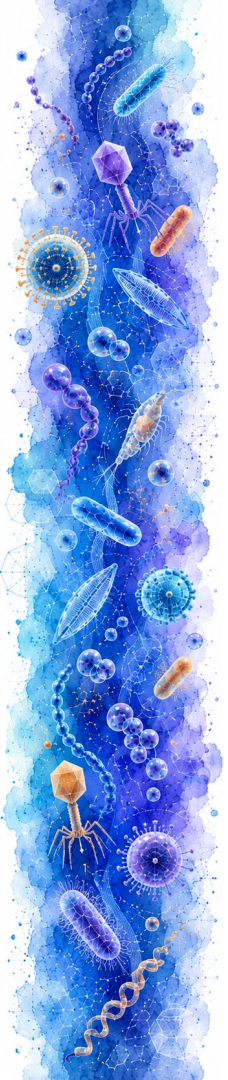
Mature biofilms show environmental gradients, for example, in oxygen and pH. Bacteria growing in the anaerobic zone in contact with the surface can use metal compounds from the material as resources, for example, through extracellular electron transfer. Extracellular reduction of metal ions, used as terminal electron acceptors, is required by certain metal-respiring bacteria. Thus, free electrons can be released and subsequently used by the bacteria for respiration, which can lead to microbially influenced corrosion. Common bacteria in metal corrosive biofilms include sulfur-redusers belonging to the **Deltaproteobacteria** (including Desulfovibrio, Desulfobacter, Desulfuromonas species), iron-oxidizers belonging to the **Gammaproteobacteria**, such as *Acidithiobacillus ferrooxidans*.





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# Biofouling Impact on Aquaculture

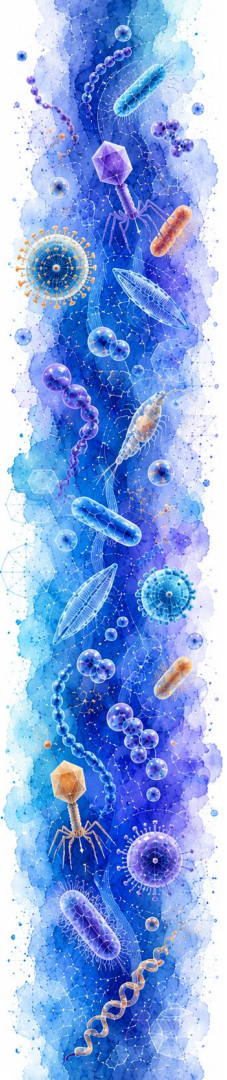
Besides indirect repercussion from increased energy consumption necessary to overcome increased frictional drag and heat and mass transfer limitations, biofouling organisms reduce water flow and increase biodeposition beneath aquaculture farms, and may be fish pathogens.

Table 2. An overview of common fouling organisms in marine finfish culture and their documented adverse impacts.

| Fouling organism | Range of known impacts | Region | Fish species affected | Author(s) |
|---|---|---|---|---|
| Chordata: Ascidiacea<br>*Ascidiella aspersa*<br>*Botrylloides* sp.<br>*Botryllus schlosseri*<br>*Styela plicata*<br>*Symplegma* sp.<br>*Tridemnum* sp. | Cage deformation and structural fatigue<br>Increased disease risk | Malaysia<br>UK | *Epinephelus* sp.<br>*Lates calcarifer*<br>*Lutjanus* sp.<br>*Salmo salar*<br>*Siganus* sp. | Milne (1975b); Tan et al. (2002); Braithwaite et al. (2007) |
| Algae<br>*Antithamnion* sp.<br>*Ectocarpus* spp.<br>*Enteromorpha* spp.<br>Filamentous diatoms<br>*Gracilaria* sp.<br>*Ulva* spp. | Net occlusion<br>Restriction of water exchange<br>Poor water quality<br>Limited oxygen availability<br>Reduced waste metabolite removal<br>Cage deformation and structural fatigue | Australia<br>UK<br>USA<br>Malaysia | Epinephelus sp<br>*Lates calcarifer*<br>Lutjanus sp.<br>*Oncorhynchus tshawytscha*<br>*Salmo salar*<br>*Siganus* sp.<br>*Thunnus maccoyii* | Milne (1975a); Milne (1975b); Moring and Moring (1975); Hodson and Burke (1994); Cronin et al. (1999); Svane et al. (2006) |
| Mollusca: Bivalvia<br>*Crassostrea* spp.<br>*Electroma georgiana*<br>*Modiolus* sp.<br>*Mytilus edulis*<br>*Perna viridis*<br>*Pinctada* spp. | Net occlusion<br>Cage deformation and structural fatigue | Australia<br>Malaysia<br>Singapore<br>UK<br>USA | *Epinephelus* sp<br>*Lates calcarifer*<br>*Lutjanus* sp.<br>*Oncorhynchus tshawytscha*<br>*Salmo salar*<br>*Siganus* sp.<br>*Thunnus maccoyii* | Milne (1975a); Milne (1975b); Moring and Moring (1975); Lee et al. (1985); Cronin et al. (1999); Braithwaite et al. (2007); Greene and Grizzle (2007) |
| Cnidaria;: Hydrozoa<br>*Ectopleura larynx*<br>*Obelia dichotoma*<br>*Plumularia* sp.<br>*Tubularia* sp. | Net occlusion<br>Reduced water flow | Malaysia<br>Norway<br>USA | *Lates calcarifer*<br>*Salmo salar* | Hodson et al. (2000); Guenther et al. (2009); Madin et al. (2009); Guenther et al. (2010); Carl et al. (2011); Guenther et al. (2011) |

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10.1080/08927014.2012.700478



# This Week Read

Qian, PY., Cheng, A., Wang, R. et al. Marine biofilms: diversity, interactions and biofouling. Nat Rev Microbiol 20, 671–684 (2022). https://doi.org/10.1038/s41579-022-00744-7

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